

Advanced (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = 3x$, what is $(f + g)(x)$?
(A) $x^2 + 3x$
(B) $x^2 - 3x$
(C) $3x^2$
(D) $x^2 \cdot 3x$
(E) $x^2/3x$
- Q2.** Given $f(x) = x - 2$ and $g(x) = x + 1$, what is $(f \cdot g)(x)$?
(A) $x^2 - x - 2$
(B) $x^2 + x - 2$
(C) $x^2 - 3$
(D) $x^2 + 1$
(E) $x^2 - 1$
- Q3.** If $f(x) = 2x$ and $g(x) = x - 1$, what is $(f/g)(x)$?
(A) $2/(x - 1)$
(B) $2(x - 1)$
(C) $2x/(x - 1)$
(D) $x - 1$
(E) $x + 1$
- Q4.** Given $f(x) = x^2$ and $g(x) = 2x - 1$, what is the domain of $(f + g)(x)$?
(A) $(-\infty, \infty)$
(B) $(-\infty, 1)$
(C) $(-1, \infty)$
(D) $[0, \infty)$
(E) $(-\infty, 0)$
- Q5.** If $f(x) = x^2 + 1$ and $g(x) = x - 2$, what is $(f \cdot g)(-1)$?
(A) -2
(B) 0
(C) 2
(D) 4
(E) 6
- Q6.** Given $f(x) = x + 2$ and $g(x) = 2x - 1$, what is $(f/g)(0)$?
(A) -1
(B) 0
(C) 1
(D) 2
(E) undefined
- Q7.** If $f(x) = x^2$ and $g(x) = x + 1$, what is $(f/g)(x)$?
(A) $x - 1$
(B) $x + 1$
(C) $x^2/(x + 1)$
(D) $x/(x + 1)$
(E) $x^2 - 1$
- Q8.** Given $f(x) = 3x$ and $g(x) = x - 2$, what is $(f + g)(2)$?
(A) 6
(B) 8
(C) 10
(D) 12
(E) 4

Open Ended Questions (FRQ)

Q9. If $f(x) = x^2 + 2x - 3$ and $g(x) = x - 1$, find $(f \cdot g)(x)$. Simplify your answer.

Q10. Given $f(x) = 2x + 1$ and $g(x) = x^2 - 2x - 1$, find the domain of $(f/g)(x)$. Explain your reasoning.

Chef's Tips

- Make sure students show their work for FRQs.
- Check for simplification of expressions.
- Remind students to consider the domain of combined functions.